

```

mysql> -- Introduction to Triggers
mysql> -----
mysql> /*A trigger is a special type of stored program that automatically executes
(fires) in response to a specific
/*> event ? INSERT, UPDATE, or DELETE ? occurring on a specified table. Unlike a
stored procedure
/*> a trigger is never called directly.
/*>
/*> General Syntax
/*> -----
/*> DELIMITER //
/*> CREATE TRIGGER trigger_name
/*> {BEFORE | AFTER} {INSERT | UPDATE | DELETE}
/*> ON table_name
/*> FOR EACH ROW
/*>
/*> BEGIN
/*>     -- trigger logic
/*>     - use NEW.column_name for inserted/updated values
/*>     -- use OLD.column_name for deleted/previous values
/*> END //
/*>
/*> DELIMITER ;
/*> */

```

```
mysql> USE COMPANYDB;
```

Database changed

```

mysql> CREATE TABLE Employees (
->     emp_id INT PRIMARY KEY,
->     emp_name VARCHAR(50),
->     department VARCHAR(30),
->     salary DECIMAL(10,2),
->     experience INT,
->     city VARCHAR(30)
-> );

```

Query OK, 0 rows affected (0.03 sec)

```
mysql>
```

```

mysql> INSERT INTO Employees VALUES
-> (101,'Rahul','IT',65000,5,'Hyderabad'),
-> (102,'Priya','HR',45000,3,'Chennai'),
-> (103,'Amit','Finance',70000,6,'Bangalore'),
-> (104,'Sneha','IT',80000,8,'Hyderabad'),
-> (105,'Kiran','Sales',40000,2,'Pune');

```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM EMPLOYEES;
```

emp_id	emp_name	department	salary	experience	city
101	Rahul	IT	65000	5	Hyderabad
102	Priya	HR	45000	3	Chennai
103	Amit	Finance	70000	6	Bangalore
104	Sneha	IT	80000	8	Hyderabad
105	Kiran	Sales	40000	2	Pune

101	Rahul	IT	65000.00	5	Hyderabad
102	Priya	HR	45000.00	3	Chennai
103	Amit	Finance	70000.00	6	Bangalore
104	Sneha	IT	80000.00	8	Hyderabad
105	Kiran	Sales	40000.00	2	Pune

5 rows in set (0.00 sec)

```
mysql> CREATE TABLE Employee_Audit(
->     audit_id INT AUTO_INCREMENT PRIMARY KEY,
->     emp_id INT,
->     action_type VARCHAR(20),
->     old_salary DECIMAL(10,2),
->     new_salary DECIMAL(10,2),
->     action_time DATETIME
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> -- BEFORE INSERT
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_before_insert_salary
-> BEFORE INSERT
-> ON Employees
-> FOR EACH ROW
-> BEGIN
->     IF NEW.salary < 20000 THEN SIGNAL SQLSTATE '45000'
-> SET MESSAGE_TEXT='Salary cannot be less than 20000';
-> END IF;
-> END//
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
mysql> INSERT INTO EMPLOYEES
-> VALUES(106,'Praveen','IT',19000,2,'Hyderabad');
ERROR 1644 (45000): Salary cannot be less than 20000
```

```
mysql>
mysql>
mysql>
mysql> -- AFTER INSERT
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_after_insert
-> AFTER INSERT
-> ON Employees
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO Employee_Audit
->     (emp_id,action_type,new_salary,action_time)
->     VALUES
->     (NEW.emp_id,'INSERT',NEW.salary,NOW());
-> END//
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
mysql> INSERT INTO EMPLOYEES
  -> VALUES(107,'Kavya','HR',55000,4,'Chennai');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM EMPLOYEE_AUDIT;
+-----+-----+-----+-----+-----+-----+
| audit_id | emp_id | action_type | old_salary | new_salary | action_time |
+-----+-----+-----+-----+-----+-----+
| 1 | 107 | INSERT | NULL | 55000.00 | 2026-09-24 14:40:34 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql> -- BEFORE UPDATE
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_before_update_salary
  -> BEFORE UPDATE
  -> ON Employees
  -> FOR EACH ROW
  -> BEGIN
  ->     IF NEW.salary<OLD.salary THEN SIGNAL SQLSTATE '45000'
  ->     SET MESSAGE_TEXT='Salary cannot be reduced';
  -> END IF;
  -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;
mysql> UPDATE EMPLOYEES SET SALARY=5000 WHERE EMP_ID=101;
ERROR 1644 (45000): Salary cannot be reduced
mysql>
mysql> -- AFTER UPDATE
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_after_update
  -> AFTER UPDATE
  -> ON Employees
  -> FOR EACH ROW
  -> BEGIN
  ->     INSERT INTO Employee_Audit
  ->     (emp_id,action_type,old_salary,new_salary,action_time)
  ->     VALUES
  ->     (NEW.emp_id,'UPDATE',OLD.salary,NEW.salary,NOW());
  -> END//
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;
mysql> SELECT * FROM EMPLOYEES;
+-----+-----+-----+-----+-----+-----+
| emp_id | last_name | first_name | salary | dept_name | 
```

emp_id	emp_name	department	salary	experience	city
101	Rahul	IT	65000.00	5	Hyderabad
102	Priya	HR	45000.00	3	Chennai
103	Amit	Finance	70000.00	6	Bangalore
104	Sneha	IT	80000.00	8	Hyderabad
105	Kiran	Sales	40000.00	2	Pune
107	Kavya	HR	55000.00	4	Chennai

6 rows in set (0.00 sec)

```
mysql> UPDATE EMPLOYEES SET SALARY=80000 WHERE EMP_ID=101;
```

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> SELECT * FROM EMPLOYEE_AUDIT;
```

audit_id	emp_id	action_type	old_salary	new_salary	action_time
1	107	INSERT	NULL	55000.00	2026-09-24 14:40:34
2	101	UPDATE	65000.00	80000.00	2026-09-24 15:02:53

2 rows in set (0.00 sec)

```
mysql>
```

```
mysql> -- BEFORE DELETE
```

```
mysql> DELIMITER //
```

```
mysql> CREATE TRIGGER trg_before_delete
```

```
-> BEFORE DELETE
```

```
-> ON Employees
```

```
-> FOR EACH ROW
```

```
-> BEGIN
```

```
-> IF OLD.department='IT' Then SIGNAL SQLSTATE '45000'
```

```
-> SET MESSAGE_TEXT='Cannot delete IT employees';
```

```
-> END IF;
```

```
-> END//
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
```

```
mysql> delete from employees where emp_id=101;
```

ERROR 1644 (45000): Cannot delete IT employees

```
mysql>
```

```
mysql>
```

```
mysql> -- AFTER DELETE
```

```
mysql> DELIMITER //
```

```
mysql> CREATE TRIGGER trg_after_delete
```

```
-> AFTER DELETE
```

```
-> ON Employees
```

```
-> FOR EACH ROW
```

```
-> BEGIN
```

```

-> INSERT INTO EMPLOYEE_AUDIT
-> (emp_id,action_type,old_salary,action_time)
-> VALUES
-> (OLD.emp_id,'DELETE',OLD.salary,NOW());
-> END//

```

Query OK, 0 rows affected (0.01 sec)

```

mysql> DELIMITER ;
mysql> DELETE FROM EMPLOYEES WHERE EMP_ID=105;
Query OK, 1 row affected (0.01 sec)

```

```
mysql> SELECT * FROM EMPLOYEE_AUDIT;
```

audit_id	emp_id	action_type	old_salary	new_salary	action_time
1	107	INSERT	NULL	55000.00	2026-09-24 14:40:34
2	101	UPDATE	65000.00	80000.00	2026-09-24 15:02:53
3	105	DELETE	40000.00	NULL	2026-09-24 15:13:59

3 rows in set (0.00 sec)

```
mysql> notee
```